



Project Title: “Your project title”

Project Title: "A Skills-to-Income Marketplace for Urban Youth in Rwanda"

Program Name: Bachelor of Software Engineering (BSE)

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Course: Pre-Capstone Project — Research Proposal

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Declaration

I, Adebayo Seyi, declare that this research proposal is my own original work and has not been submitted, in whole or in part, for any degree or examination at this or any other institution. All sources of information used have been duly acknowledged through complete and accurate references.

Student Signature: _____

Date: _____

This research proposal has been submitted for examination with my approval as Supervisor.

Supervisor: Hubert Apana

Signature: _____

Date: _____

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List of Acronyms/Abbreviations

ALU — African Leadership University

API — Application Programming Interface

BSE — Bachelor of Software Engineering

CI/CD — Continuous Integration / Continuous Deployment

ERD — Entity Relationship Diagram

GSMA — GSM Association (global mobile industry body)

JWT — JSON Web Token

MoMo — Mobile Money

MVP — Minimum Viable Product

NISR — National Institute of Statistics of Rwanda

ORM — Object-Relational Mapping

REST — Representational State Transfer

UI/UX — User Interface / User Experience

UML — Unified Modeling Language

UUID — Universally Unique Identifier

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Youth unemployment and underemployment remain among the most persistent socioeconomic challenges facing Rwanda and the wider East African region. According to the **National Institute of Statistics of Rwanda (NISR, 2024)**, although Rwanda has made significant progress in job creation, many young people continue to experience unemployment, underemployment, and limited access to stable, productive work. A large proportion of young people in urban centres such as Kigali possess practical, income-generating skills, including electronics and phone repair, tailoring, tutoring, and digital services such as web design, yet lack structured, trustworthy channels through which to convert these skills into consistent income. Work is typically found through word-of-mouth referrals, informal WhatsApp groups, or by waiting to be selected for casual labour, none of which allow a worker to build a verifiable, portable record of competence that a new client can trust before engaging them.

This project, **Isoko Talents**, responds directly to the author's stated mission: *“to create sustainable job opportunities for African youth through technology, innovation, and community-driven solutions that empower individuals economically and socially.”* The project proposes a mobile-responsive web platform that lets urban youth prove their skills through mentor-reviewed micro-tasks, get matched to nearby income opportunities through a geolocation- and trust-weighted matching engine, and receive guidance from experienced mentors, while a simulated mobile-money payment flow demonstrates how earnings could be tracked and eventually paid out through real providers such as MTN Mobile Money or Airtel Money. This approach aligns with Rwanda's national development agenda, which emphasizes digital innovation, youth employment, and entrepreneurship as key drivers of inclusive economic growth (**Government of Rwanda, 2020**).

The scope of this proposal is deliberately limited to the urban, skilled gig-economy segment (repair work, tailoring, tutoring, and digital services) rather than agricultural or rural labour matching, a decision explained further under **Project Scope** below. This focus also reflects recent findings from the **World Bank (2025)**, which highlight that modernizing employment systems, improving labour market efficiency, and creating more productive jobs for young people are essential to sustaining Rwanda's economic transformation.

1.2 Problem Statement

Three interrelated problems prevent Rwandan urban youth with informal or practically-acquired skills from converting those skills into stable income:

- **Invisibility of skill:** informal workers have no verifiable, portable record of competence that a prospective client can check before engaging them, forcing every new client relationship to start from zero trust.
- **Fragmented, ad-hoc discovery:** existing digital job platforms in the region concentrate on formal, white-collar hiring, or, where informal-work platforms do exist, they cover only a narrow slice of services (e.g., home repairs) or provide a profile with no functioning matching or hiring mechanism.
- **Disconnected financial rails:** even where informal work is found, payment is typically cash-based and undocumented, leaving workers without the transaction history that mobile-money-linked micro-financing increasingly depends on.

Isoko Talents addresses these problems by combining structured skill verification, hyperlocal matching, and a trackable payment record into a single platform purpose-built for the urban East African informal labour context.

1.3 Project's Main Objective

To design and develop a full-stack web platform that enables urban Rwandan youth to build mentor-verified skill profiles, discover and secure nearby gig opportunities through a trust-weighted matching engine, and track earnings through a simulated mobile-money payment workflow, thereby contributing to sustainable, community-driven economic empowerment.

1.4 List of the Specific Objectives

- To design and implement a secure, role-based authentication and profile system for workers, clients, mentors, and administrators.
- To design a category-specific skill-verification mechanism in which workers submit real micro-task evidence, reviewed by an independently assigned mentor pool, generating a portable trust score resistant to peer collusion.
- To implement a geolocation-based gig-matching engine that ranks candidate workers by a weighted combination of proximity and trust score.
- To build a simulated mobile-money payment workflow that mirrors the request/response contract of a real provider (e.g., MTN MoMo), recording a full, auditable transaction history without moving real funds.
- To build a lightweight mentorship messaging layer connecting new workers to experienced professionals.
- To containerize the application using Docker and establish a CI/CD pipeline for automated testing and deployment.
- To evaluate the resulting platform against defined technical and adoption metrics through structured testing with a representative pilot group of workers and clients.

1.5 Research Questions

- RQ1: What mechanism can reliably verify the practical skills of informal urban workers in a way that resists collusion between a worker and their reviewer?
- RQ2: How should a gig-matching algorithm weight geographic proximity against worker trustworthiness to produce fair, useful matches in a hyperlocal, informal labour market?
- RQ3: To what extent does a structured, mentor-verified skill-proof system increase a client's stated willingness to hire a previously unknown informal worker, compared with an unverified profile?

1.6 Project Scope

The Minimum Viable Product (MVP) for this capstone is deliberately bound to keep the deliverable achievable within a 16-week term and testable with real users.

In scope:

- Four urban skill categories at launch: electronics/phone repair, tailoring, tutoring, and digital/web work.
- Authentication, role-based profiles, skill-verification micro-tasks, gig posting/browsing, geolocation-based matching, ratings, and mentorship messaging.
- A simulated mobile-money payment flow, not a live financial integration.

- A mobile-responsive web interface, accessible via any modern smartphone browser, rather than a native app.

Out of scope for this capstone term:

- Live integration with a real mobile money provider, which requires a registered business account and provider approval beyond the academic timeline.
- Farm labour / agricultural gig matching, which is a related but distinct problem requiring farm partnerships and a full growing-season cycle to properly pilot; it is noted here as a possible future extension rather than part of this project.
- A native Android/iOS application and offline/USSD support for feature-phone users.
- Automated fraud-detection systems beyond basic moderation flags.

1.7 Significance and Justification

This project is significant at three levels. Practically, it offers a concrete tool that could help urban Rwandan youth convert existing skills into discoverable, trackable income, directly supporting national and regional youth-employment priorities. Academically, it contributes a specific, testable design for skill verification in low-trust, informal labour markets, a problem under-addressed in existing gig-economy literature, which tends to focus on formal-sector or platform-mediated Global North contexts. Personally, it builds directly on the author's prior technical work (IoT-based agricultural monitoring) and continued interest in African digital inclusion, demonstrated through representation at the Smart Africa conference, extending that interest into a full-stack, trust-and-livelihood-focused system.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature and platforms relevant to informal labour markets, digital gig-economy platforms, and trust/reputation mechanisms, in order to situate Isoko Talents within the existing body of work and to identify where current research and practice fall short for the specific context of urban, informal, skilled youth labour in Rwanda. The review draws on labour statistics, mobile-money adoption data, comparable regional platforms, and established theoretical work on information asymmetry and trust in markets.

2.2 Related Literature

2.2.1 Youth Unemployment and the Informal Economy in Rwanda

National labour statistics highlight youth unemployment and underemployment as a persistent structural challenge in Rwanda, with a large share of economic activity occurring in the informal sector where workers lack formal contracts, verifiable credentials, or documented income history (National Institute of Statistics of Rwanda, 2026). This body of data establishes the scale of the problem this project addresses but does not, on its own, propose a mechanism for making informal skill visible or verifiable.

2.2.2 Digital Gig-Economy Platforms and Matching Mechanisms

Globally, gig-economy platforms have demonstrated that digital matching can meaningfully reduce search friction between workers and clients. However, most prominent platforms are designed around formal, urban, often Global North contexts, with matching logic that assumes reliable digital payment rails and a baseline level of institutional trust (e.g., background checks, bank-linked identity) that does not translate directly to informal, cash-based, low-connectivity markets. Regionally, KaziLinks (Kenya) demonstrates on-demand booking for vetted home-service technicians in Nairobi, but is narrowly scoped to service booking and does not include a skill-verification or trust-scoring mechanism. A related Rwandan hackathon prototype, also named KaziLink, explored digital profiles for verifying informal workers' history, but did not progress to a functioning marketplace with matching or payment capability.

2.2.3 Trust and Reputation Systems in Peer-to-Peer Markets

A separate stream of literature addresses trust and reputation mechanisms in peer-to-peer and platform markets, generally converging on the idea that visible, verifiable signals of quality reduce information asymmetry between strangers transacting for the first time. This connects directly to Spence's (1973) foundational signaling theory, which shows that costly, hard-to-fake signals (such as a completed and independently reviewed task) can credibly convey underlying quality in a market where quality is otherwise unobservable, precisely the situation facing an informal worker meeting a new client for the first time.

2.2.4 Mobile Money and Financial Inclusion in Rwanda

Mobile money services, principally MTN Mobile Money and Airtel Money, have achieved near-ubiquitous reach across Rwanda, including rural and lower-income populations (GSMA, 2026). This establishes mobile money as a realistic payment rail for a platform like Isoko Talents, even though, as noted in the project scope, live integration with these providers is outside the current capstone's technical boundary.

2.2.5 Digital Trust Precedents in Rwandan E-Government

Rwanda's Irembo platform illustrates that digitizing access to previously informal or bureaucratic processes can meaningfully build public trust in digital systems when the platform is reliable and easy to use. While Irembo addresses government services rather than labour markets, its adoption pattern is a useful local precedent for the kind of trust-building this project also depends on.

2.3 Research Gap

Across the literature and comparable platforms reviewed, three gaps recur, none of which are addressed together by any single existing source or system:

- **No combined verification-plus-matching system for informal urban work:** existing regional platforms address either service booking (KaziLinks) or worker profiling (KaziLink prototype), but not both together with a live matching engine.
- **Limited application of signaling theory to informal, gig-based labour:** signaling theory is well-established in formal labour-market economics but is rarely operationalized as a concrete product mechanism for informal, skills-based gig work in an African urban context.
- **Under-specification of anti-collusion design in peer-verification systems:** where skill-verification concepts do appear in early-stage regional prototypes, the literature and existing systems reviewed do not specify how reviewer assignment and reviewer accountability should be structured to resist gaming.

Isoko Talents is positioned to directly address these three gaps through its category-specific, mentor-reviewed skill-verification mechanism with independent reviewer assignment, combined with a live, trust-weighted matching engine.

2.4 Conceptual Framework

The project's design is grounded in two complementary theoretical perspectives:

- **Signaling theory (Spence, 1973):** in a market where a worker's true skill quality is unobservable to a new client, a costly, hard-to-fake signal, here, a real micro-task independently reviewed by an assigned mentor, credibly conveys quality and reduces the information asymmetry that otherwise prevents a first transaction between strangers.
- **Social embeddedness of informal labour markets (Granovetter, 1973):** informal work has historically been coordinated through personal networks and “weak ties” rather than formal institutions; the platform's mentorship layer and hyperlocal matching are designed to digitize and extend these existing weak-tie dynamics rather than replace them with an impersonal formal system.

Together, these perspectives inform the platform's core causal chain: Skill Verification (a credible signal) → Trust Score (an accumulated, portable reputation) → Matching (connecting the signal to relevant nearby demand) → Income Outcome (the practical result the project's mission targets).

2.5 Conclusion

The reviewed literature confirms that youth unemployment and informal-sector invisibility are well-documented structural problems in Rwanda, that mobile money provides a realistic underlying payment infrastructure, and that signaling and social-network theory offer a credible theoretical basis for a skill-verification and matching mechanism. However, no existing regional platform combines verified skill signaling with live, trust-weighted, hyperlocal matching for urban informal work. This capstone project's remaining chapters, particularly the system design in Chapter Three, directly target this gap as the area for further applied research and system-building.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the research design and system-development methodology used to build and evaluate Isoko Talents. The project combines an iterative Agile software-development approach with a structured system-analysis and design process, followed by pilot evaluation against defined technical and adoption metrics. Each stage is described briefly and concisely below, with supporting diagrams and requirement tables.

3.2 Research Design

The project follows an Agile, iterative development model structured around two-week sprints, each delivering a testable increment of functionality accompanied by unit and integration tests. A lightweight Scrum approach is used: sprint planning against a prioritized product backlog, active development and testing, a sprint review with the supervisor, and a retrospective that feeds directly back into the backlog for the next sprint. This model was chosen because it allows the system's riskiest and most novel component, the skill-verification and matching logic, to be validated early through working software rather than left until a single late-stage integration.



Figure 1: Proposed Development Model (Agile/Scrum)

3.3 Dataset Description and Presentation

Not applicable. This capstone project follows the Full-Stack Development specialization track rather than the Machine Learning specialization track, and therefore does not involve a trained model or a curated dataset in the sense required by this section. The project's data model is instead presented as a relational database schema in Section 3.5 (System Design).

3.3 System Analysis

3.3.1 Analysis of Existing Systems

Three existing systems were reviewed for relevant features prior to designing Isoko Talents:

System	Key Features	Limitations Relative to This Project
KaziLinks (Kenya)	On-demand booking of vetted home-service technicians (plumbing, electrical, cleaning) in Nairobi.	Limited to home-service booking; no skill-verification or trust-scoring mechanism; not built for the Rwandan context.
KaziLink (Rwanda hackathon prototype)	Digital profiles intended to verify informal workers' history and skills.	Prototype stage only; no live gig marketplace, matching engine, or payment workflow.
iSOKO (TradeMark Africa)	Regional cross-border trade platform for buying/selling goods and accessing market information.	Oriented toward goods trade, not labour/gig matching; no individual skill-verification mechanism.

Table 3: Comparison of Existing Systems

Description of the new system: Isoko Talents is organized into eight core modules, summarized below, each mapped to a corresponding data model and REST API surface.

Module	Description
Authentication & Profiles	JWT-based authentication with role-based access (worker, client, mentor, admin); profile includes bio, location, and skills.
Skill Verification	Category-specific micro-task submission, reviewed by an independently assigned mentor; generates a portable, per-skill trust score.
Gig Marketplace & Matching	Clients post gigs with location, budget, and required skill category; a matching engine ranks candidate workers by proximity and trust score.
Payments (Simulated)	Mock mobile-money transaction flow recording initiation, confirmation, and history for each completed gig.
Mentorship Messaging	In-app messaging connecting new workers with mentors for guidance.
Ratings & Reviews	Two-way rating between workers and clients after gig completion, feeding into the trust score.
Admin Moderation	Tools for reviewing flagged accounts, disputed gigs, and platform-wide activity.

3.3.2 Analysis of Current Systems

Summary of the needs expressed by the proposed system: the platform must support four concurrent user roles, a category-specific verification workflow, a location-aware matching computation, and a fully recorded (simulated) payment lifecycle, while remaining lightweight enough to deploy and demonstrate within a single containerized environment for the purposes of this capstone.

ID	Functional Requirement
FR-1	The system shall allow a user to register and authenticate using email and password, receiving a signed JWT session token.
FR-2	The system shall allow a worker to submit a category-specific skill-verification task with supporting evidence (file/link and notes).
FR-3	The system shall automatically assign a submitted skill task to an available mentor in the matching category, without allowing the worker to select their own reviewer.
FR-4	The system shall allow a client to post a gig with a title, description, budget, required skill category, and location.
FR-5	The system shall rank candidate workers for a gig by a weighted combination of distance and trust score, and return the ranked list to the client.
FR-6	The system shall allow a worker or client to update the status of a match (pending, accepted, completed, cancelled).
FR-7	The system shall generate a simulated payment transaction once a match is marked completed, and record its status (initiated, confirmed, failed).
FR-8	The system shall allow both parties in a completed match to leave a rating and comment, which updates the recipient's trust score.
FR-9	The system shall allow direct messaging between a worker and an assigned mentor.

Table 1: Functional Requirements

ID	Non-Functional Requirement
NFR-1	Usability: the interface shall be mobile-responsive and usable on common Android smartphone browsers, given the platform's mobile-first user base.
NFR-2	Performance: standard API endpoints (excluding the matching computation) shall respond within 500ms under normal test load.
NFR-3	Security: passwords shall be hashed (bcrypt) and all protected routes shall require a valid JWT; sensitive fields shall never be returned in API responses.
NFR-4	Reliability: the CI/CD pipeline shall run automated tests on every commit before any deployment build proceeds.
NFR-5	Maintainability: the codebase shall follow a layered architecture (routes → controllers → services → models) to keep components independently testable.

NFR-6	Data Privacy: personal and location data shall be handled in line with Rwanda's Law No. 058/2021 on the protection of personal data and privacy.
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Table 2: Non-Functional Requirements

3.4 System Architecture

The system follows a three-tier architecture: a React single-page application (client tier), a Node.js/Express REST API (application tier), and a PostgreSQL relational database (data tier). Authentication is handled via JSON Web Tokens; the matching engine and payment simulation run as service-layer modules within the API. The entire stack is containerized with Docker Compose and deployed through a CI/CD pipeline that runs automated tests on every push before building and publishing container images.

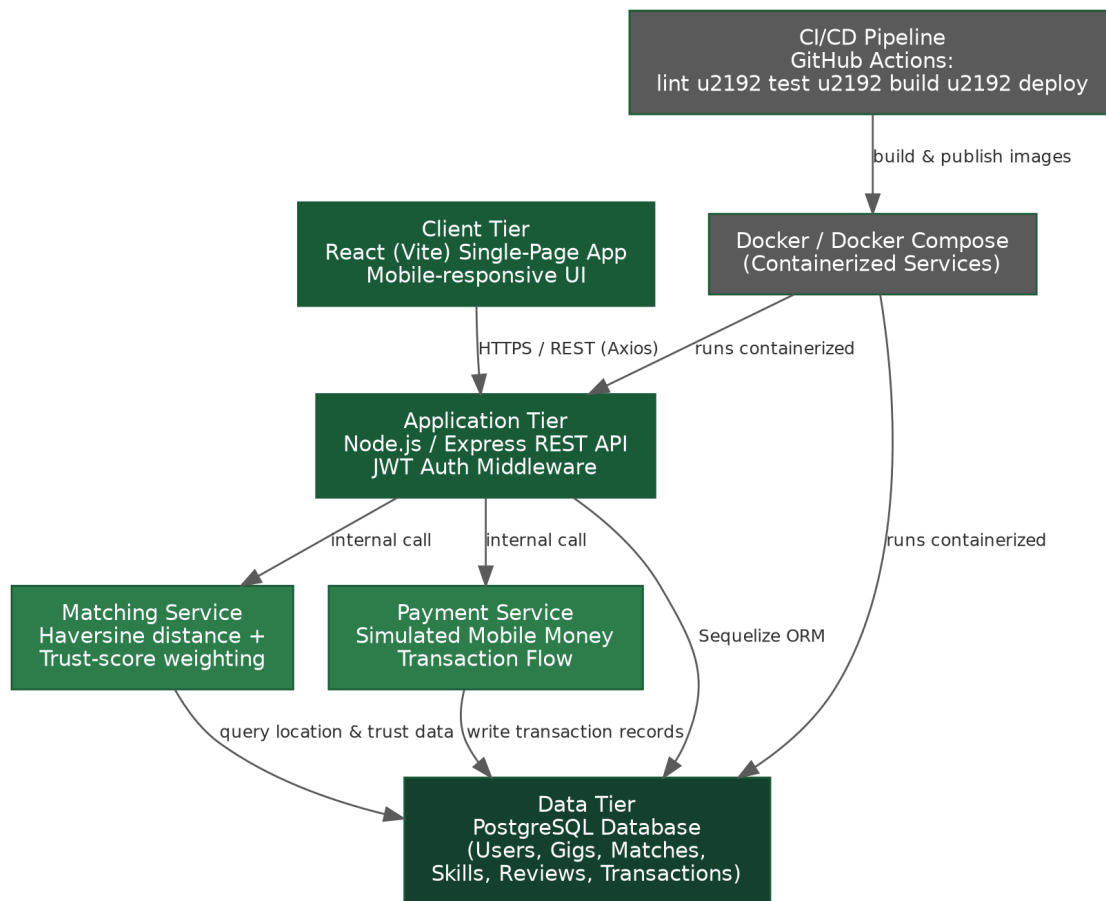


Figure 2: System Architecture Diagram

Interpretation: client requests flow through the Express API, which delegates location- and trust-weighted ranking to the Matching Service and payment lifecycle handling to the Payment Service, both of which read and write through a single Sequelize ORM layer to PostgreSQL. This separation keeps the matching and payment logic independently testable and replaceable, for example, swapping the simulated Payment Service for a real MTN MoMo integration in a future phase, without altering the client tier or the rest of the API.

3.5 System Design

This section presents the mandatory system design diagrams: the Entity Relationship Diagram, the Class Diagram, and the Use Case Diagram.

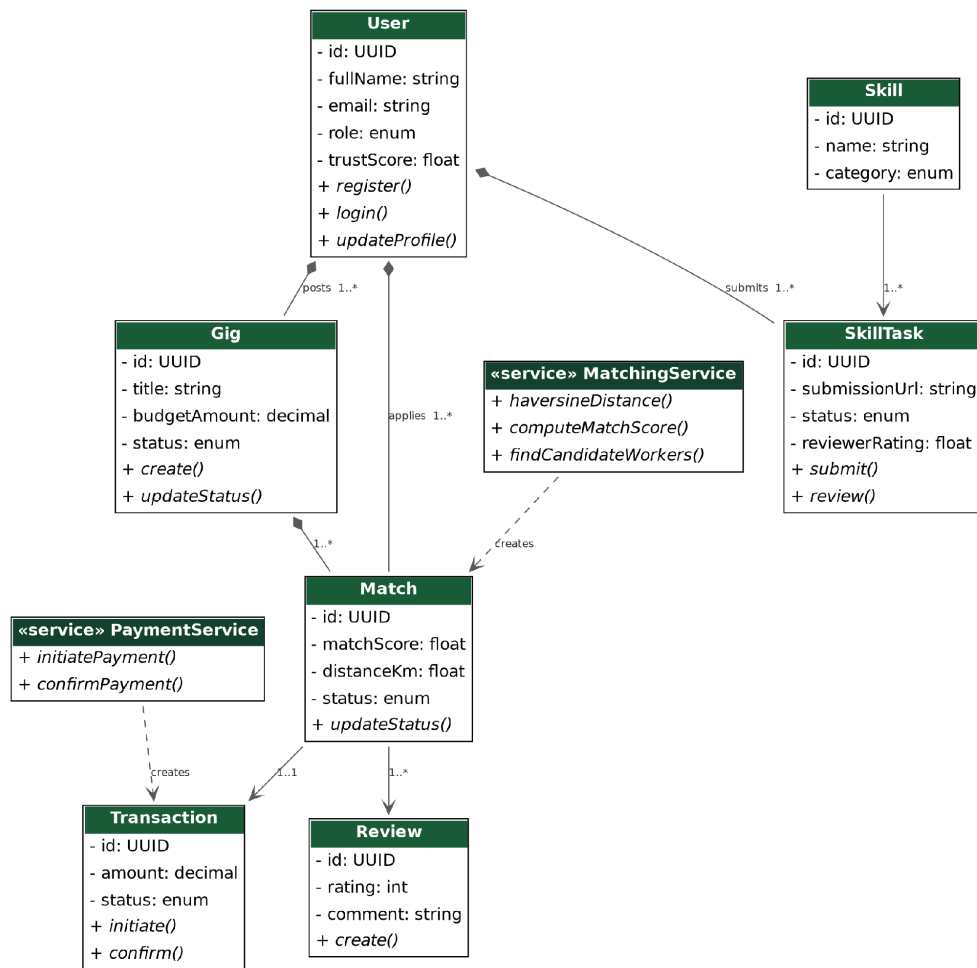


Figure 4: Class Diagram

Interpretation: the class diagram mirrors the ERD's core entities but adds behaviour, and introduces two stereotyped service classes, MatchingService and PaymentService, which encapsulate the ranking computation and simulated payment lifecycle respectively, keeping that logic out of the data model classes themselves.

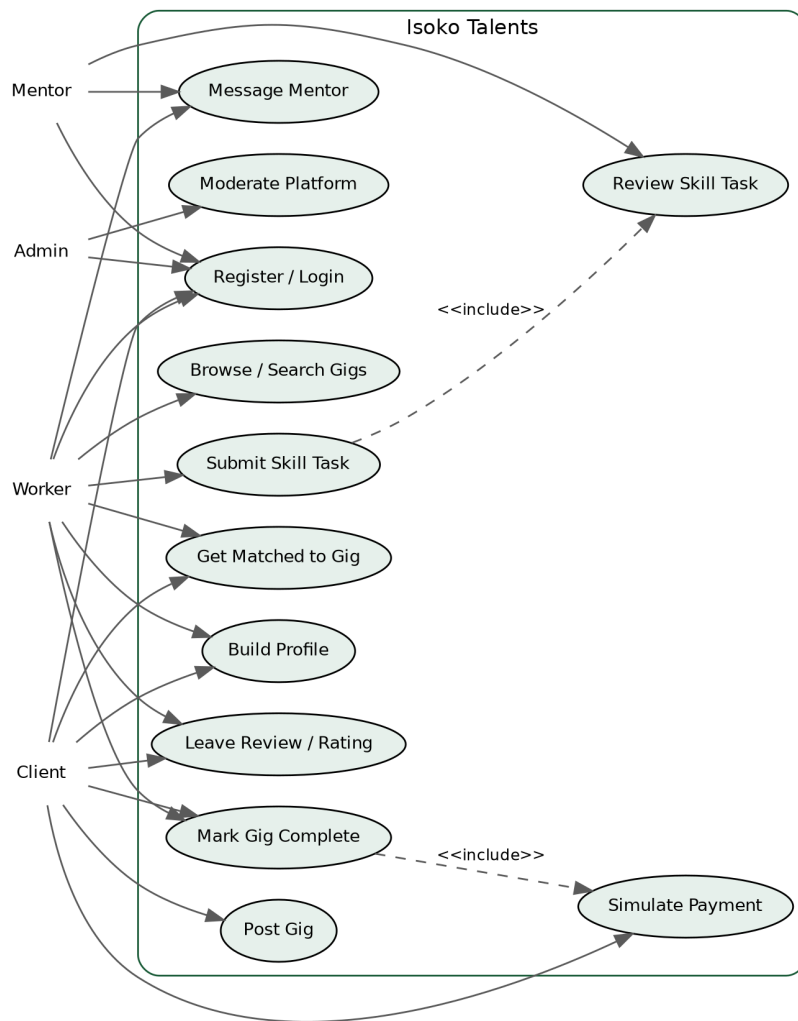


Figure 5: Use Case Diagram

Interpretation: four actors, Worker, Client, Mentor, and Admin, interact with a shared set of use cases. Submitting a skill task includes reviewing it (restricted to the Mentor actor), and marking a gig complete includes triggering the simulated payment, reflecting the automatic, non-optional sequencing built into the platform's workflow.

3.6 Development Tools

Tool / Technology	Purpose
React (Vite)	Frontend single-page application framework.
Node.js / Express	Backend REST API framework.
PostgreSQL	Relational database for users, gigs, matches, and transactions.
Sequelize	ORM for schema migrations and model relationships.
JWT + bcrypt	Stateless authentication and secure password hashing.
Docker / Docker Compose	Containerization for consistent development and deployment environments.
GitHub Actions	CI/CD pipeline for automated testing and build/deploy steps.
Graphviz	Generation of system diagrams (ERD, class diagram, use case diagram, architecture).
Postman	Manual and automated API endpoint testing during development.

Visual Studio Code	Primary development environment.
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Table 4: Development Tools and Justification

3.7 Ethical Considerations

- **Data privacy:** personal data (identity, location, contact details) will be collected and stored in line with Rwanda's Law No. 058/2021 relating to the protection of personal data and privacy; no personal data will be shared with third parties outside the scope of this capstone.
- **Informed consent for pilot testing:** all pilot participants (workers, clients, and mentors) will be informed of the platform's academic, non-commercial capstone status, and that payment within the pilot is simulated and does not involve real funds.
- **No real financial risk:** because the payment workflow is simulated for this capstone, no pilot participant is exposed to financial loss through the payment feature itself.
- **Fair and auditable reviewing:** the mentor-assignment mechanism (Section 3.3.1) is designed specifically to prevent a worker and reviewer from colluding, and mentor review histories will be auditable by the project administrator during the pilot.
- **Voluntary participation and withdrawal:** pilot participants may withdraw their data and participation at any point without penalty.
- **Academic integrity:** all third-party code libraries, design references, and data sources used in this project are credited in the References section.

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